Deep Learning for Economics and Finance

HJB, Kolmogorov forward, Aiyagari in continuous time

Simon Scheidegger

HEC, University of Lausanne

Based on: Achdou, Han, Lasry, Lions & Moll (2022), *Rev. Econ. Stud.* 89(1), 45–86.

Gu, Lauriere, Merkel & Payne (2024), "Global Solutions to Master Equations." arXiv:2406.13726.

Payne, J. (2023–2024), ECO 529 Lecture Notes, Princeton University.

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Where We Are in This Course

Block 1 (PINNs):

- Single PDEs: Poisson, HJB, Black–Scholes
- Neural network $\approx$ PDE solution
- Residual loss + automatic differentiation

Block 2 (this block):

- Coupled PDEs: HJB + KFE
- PDEs arise from economic equilibrium
- Continuum of heterogeneous agents

From PINNs to Heterogeneous Agent Models

Block 1: PINNs
Solve *one* PDE
$\mathcal{L}[u] = 0$ on Ω

$\xrightarrow{\text{extend}}$

Block 2: Het. Agents
Solve *coupled* system
HJB + KFE + Market Clearing

Recall from Deck 1: the cake-eating HJB you just solved with a PINN is the single-PDE special case (one agent, no income, no distribution). We now *add* idiosyncratic income, which forces a population density g into the picture and brings the KFE alongside the HJB.

The key insight: in continuous-time macro with heterogeneous agents, equilibrium is characterized by a system of PDEs:

1. **HJB equation** (backward): each agent optimizes $\Rightarrow$ value function V, policy c^*
2. **Kolmogorov forward equation** (forward): population density g evolves given c^*
3. **Market clearing**: aggregate supply = aggregate demand $\Rightarrow$ prices r, w

Why Continuous Time?

Advantages of continuous-time formulation:

- Analytical tractability: Ito calculus provides closed-form FOCs
- Clean separation: HJB (backward) $\leftrightarrow$ KFE (forward)
- Powerful numerical methods: finite differences, PINNs
- No expectations operator: replaced by differential operators
- Connection to mean field games (Lasry & Lions, 2007)

Key References

- Achdou, Han, Lasry, Lions & Moll (2022), *REStud*
- Kaplan, Moll & Violante (2018), *AER*
- Moll lecture notes: `benjaminmoll.com/lectures/`

Discrete $\rightarrow$ Continuous

The DEQN/Young's method block used *discrete time*.
Today: *continuous time* $\Rightarrow$ PDEs.

The Aiyagari–Bewley–Huggett Framework

- Benchmark models in "quantitative" macroeconomics:
  - **Bewley** (1980, 1983): precautionary saving with borrowing constraints
  - **Huggett** (1993, *JEDC*): endowment economy, risk-free rate determination
  - **Aiyagari** (1994, *QJE*): production economy, general equilibrium
  - **Krusell & Smith** (1998, *JPE*): adds aggregate TFP shocks
- Core mechanism: idiosyncratic labor income risk is uninsurable
  - Households want to borrow when income is low $\rightarrow$ borrowing limits prevent this
  - Instead, households accumulate assets as precautionary savings
- These are building blocks for HANK models (Kaplan, Moll & Violante, 2018)
  - Heterogeneous Agent New Keynesian: monetary/fiscal policy with inequality

Goals for This Lecture

1. **Stochastic calculus refresher**: Brownian motion, Ito's lemma, Poisson processes
2. **Derive the Kolmogorov forward equation** (KFE / Fokker–Planck):
   - First as a standalone concept from physics (diffusion of particles)
   - Then in economics: continuum of agents $\rightarrow$ wealth distribution
3. **The Hamilton–Jacobi–Bellman equation**: stochastic optimal control in CT
4. **Competitive equilibrium**: coupled HJB + KFE + market clearing
5. **The master equation**: collapsing the system into a single PDE

Companion deck (numerical methods): finite differences, PINNs, and deep learning (EMINNs) for solving these coupled PDEs.

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Stochastic Calculus Refresher

The Kolmogorov Forward Equation

The HJB Equation

Competitive Equilibrium

The Master Equation

Learning objectives

By the end of this lecture you should be able to:

- Derive the Hamilton–Jacobi–Bellman equation from stochastic optimal-control problems with idiosyncratic risk
- Construct and solve the Kolmogorov forward (Fokker–Planck) equation for wealth-distribution dynamics given optimal policies
- Recognize how continuous-time stochastic calculus (Brownian motion, Ito's lemma, Poisson jumps) replaces discrete expectation operators
- Assemble coupled HJB + KFE + market-clearing systems to define competitive equilibrium in heterogeneous-agent economies (Huggett, Aiyagari, Krusell–Smith)
- Reformulate a coupled PDE system as a single master equation and interpret its solution structure

Key References

Foundational models:

- Huggett, M. (1993). "The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies." *JEDC* 17(5–6), 953–969.
- Aiyagari, S.R. (1994). "Uninsured Idiosyncratic Risk and Aggregate Saving." *QJE* 109(3), 659–684.
- Krusell, P. & Smith, A.A. (1998). "Income and Wealth Heterogeneity in the Macroeconomy." *JPE* 106(5), 867–896.

Continuous-time formulations:

- Achdou, Han, Lasry, Lions & Moll (2022). "Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach." *REStud* 89(1), 45–86.
- Gu, Lauriere, Merkel & Payne (2024). "Global Solutions to Master Equations for CT Het. Agent Models." arXiv:2406.13726.
- Lasry & Lions (2007). "Mean Field Games." *Jpn. J. Math.* 2(1), 229–260.

Part I

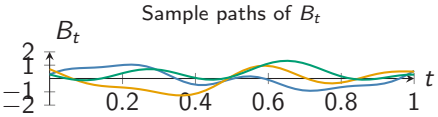
Stochastic Calculus Refresher

Brownian Motion (Wiener Process)

Definition

A stochastic process $\{B_t\}_{t \geq 0}$ is a standard Brownian motion if:

1. $B_0 = 0$
2. **Independent increments**: $B_t - B_s \perp B_s - B_r$ for $r < s < t$
3. **Gaussian**: $B_t - B_s \sim \mathcal{N}(0, t - s)$
4. **Continuous paths**: $t \mapsto B_t$ is continuous a.s.



Key properties:

- $\mathbb{E}[B_t] = 0, \quad \mathrm{Var}(B_t) = t$
- Paths are nowhere differentiable
- Quadratic variation: $\langle B \rangle_t = t$

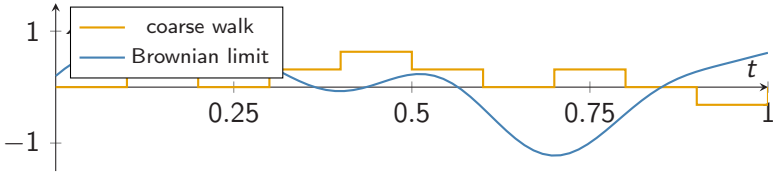
Brownian Motion: Random Walk Limit

Discrete-time intuition: Brownian motion is the limit of a scaled random walk.

Let $\Delta t > 0$ and define $X_t^{\Delta t}$ by steps $\pm\sqrt{\Delta t}$ with equal probability:

$$X_{t+\Delta t}^{\Delta t} = X_t^{\Delta t} + \sqrt{\Delta t}\,\varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \{-1, +1\} \text{ with prob. } \tfrac{1}{2} \text{ each.}$$

As $\Delta t \to 0$: $\quad X_t^{\Delta t} \xrightarrow{\text{d}} B_t \qquad$ (Donsker's theorem).



The $\sqrt{\Delta t}$ scaling is crucial: it ensures $\mathrm{Var}(X_t) = t$, not 0 or ∞.

Itô Processes and SDEs

General Itô Process

A stochastic process X_t follows an Itô process if:

$$\boxed{dX_t = \mu(X_t, t)\,\mathrm{d}t + \sigma(X_t, t)\,\mathrm{d}B_t}$$

- $\mu(X_t, t)$: drift (deterministic trend)
- $\sigma(X_t, t)$: diffusion coefficient (volatility / noise intensity)
- B_t: standard Brownian motion

Important examples:

Geometric Brownian Motion (GBM)

$dS_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}B_t$

Used for: stock prices, GDP

Ornstein–Uhlenbeck (OU) Process

$dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$

Used for: productivity, interest rates

The Ornstein–Uhlenbeck Process

$$dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$$

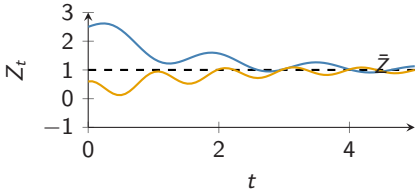
Properties:

- Mean-reverting: Z_t is pulled toward $\bar{Z}$ at rate η
- Stationary distribution: $Z_\infty \sim \mathcal{N}\left(\bar{Z},\ \frac{\sigma^2}{2\eta}\right)$
- Closed-form solution by integrating factor.

In this lecture:

- Aggregate TFP: $dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t^0$
- Krusell–Smith (1998): Z_t drives aggregate fluctuations

OU sample paths ($\bar{Z} = 1$)



Poisson Process and Continuous-Time Markov Chains

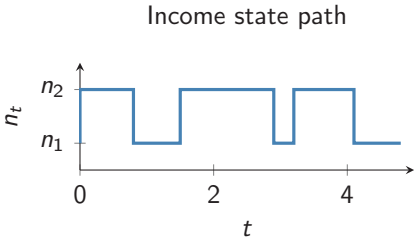
Poisson Process

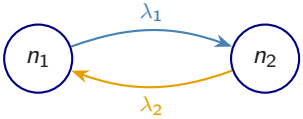
N_t: counting process with intensity $\lambda > 0$.

- Jumps of size $+1$ at random times
- Inter-arrival times $\sim \text{Exp}(\lambda)$
- $\Pr(\text{jump in } [t, t+h)) = \lambda h + o(h)$

Income switching (for our models):

- Labor state $n_t \in \{n_1, n_2\}$ (e.g., employed/unemployed)
- $n_1 \to n_2$ with intensity λ_1
- $n_2 \to n_1$ with intensity λ_2
- Ergodic probabilities: $\pi_1 = \frac{\lambda_2}{\lambda_1+\lambda_2}$, $\pi_2 = \frac{\lambda_1}{\lambda_1+\lambda_2}$





Itô's Lemma: The Chain Rule of Stochastic Calculus

Itô's Lemma

Let X_t follow $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$ and let $f \in C^2$. Then:

$$\boxed{df(X_t) = \left[f'(X_t)\,\mu + \tfrac{1}{2}f''(X_t)\,\sigma^2\right]\,\mathrm{d}t + f'(X_t)\,\sigma\,\mathrm{d}B_t}$$

The key difference from ordinary calculus:

- In ordinary calculus: $df = f'\,dX$ (first-order only)
- In stochastic calculus: $df = f'\,dX + \frac{1}{2}f''\,(dX)^2$ (second-order term survives!)
- Why? Because $(dB_t)^2 = dt \neq 0$ (Brownian motion has "infinite" variation)

Differential algebra rules:

$$\mathrm{d}t \cdot \mathrm{d}t = 0, \qquad \mathrm{d}t \cdot \mathrm{d}B_t = 0, \qquad \mathrm{d}B_t \cdot \mathrm{d}B_t = \mathrm{d}t$$

Itô's Lemma: Worked Example

Example: Let $dX_t = \mu X_t \, \mathrm{d}t + \sigma X_t \, \mathrm{d}B_t$ (GBM). Find $d(\ln X_t)$.

Step 1: Let $f(x) = \ln x$, so $f'(x) = 1/x$, $f''(x) = -1/x^2$.

Step 2: Apply Itô's Lemma with $\mu(X_t) = \mu X_t$, $\sigma(X_t) = \sigma X_t$:

$$d(\ln X_t) = \frac{1}{X_t} \cdot \mu X_t \, \mathrm{d}t + \frac{1}{2}\left(-\frac{1}{X_t^2}\right)(\sigma X_t)^2 \, \mathrm{d}t + \frac{1}{X_t} \cdot \sigma X_t \, \mathrm{d}B_t$$

$$= \boxed{\left(\mu - \tfrac{1}{2}\sigma^2\right) \, \mathrm{d}t + \sigma \, \mathrm{d}B_t}$$

Insight: $\ln X_t$ is an Itô process with *constant* drift $\mu - \frac{1}{2}\sigma^2$ and diffusion σ.

$\Rightarrow$ Solution: $X_t = X_0 \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma B_t\right]$

The $-\frac{1}{2}\sigma^2$ Itô correction is why expected return $\neq$ log return for volatile assets.

Itô's Lemma: Multivariate Version

For $V(X_t, t)$ where $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$:

$$\boxed{dV = \left[\frac{\partial V}{\partial t} + \mu\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 V}{\partial x^2}\right]\mathrm{d}t + \sigma\frac{\partial V}{\partial x}\,\mathrm{d}B_t}$$

With jumps (Poisson process with intensity λ, jump ΔX):

$$dV = \underbrace{\left[\frac{\partial V}{\partial t} + \mu\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 V}{\partial x^2}\right]\mathrm{d}t + \sigma\frac{\partial V}{\partial x}\,\mathrm{d}B_t}_{\text{diffusion part}} + \underbrace{\left[V(X_{t^-} + \Delta X, t) - V(X_{t^-}, t)\right]\mathrm{d}N_t}_{\text{jump part}}$$

We will use the multivariate version repeatedly:
HJB derivation (Part IV) and **KFE derivation** (Part II).

Part II

The Kolmogorov Forward Equation

(Fokker–Planck Equation)

What Is the Kolmogorov Forward Equation?

The Kolmogorov forward equation (KFE), also known as the Fokker–Planck equation, describes how the **probability density** of a stochastic process evolves over time.

Origin: physics and applied mathematics.

- **A.N. Kolmogorov** (1931): general theory of continuous Markov processes
- **A.D. Fokker** (1914) & **M. Planck** (1917): diffusion of particles
- Applications: heat conduction, fluid dynamics, plasma physics, chemical kinetics, population genetics, neuroscience, ...

A.N. Kolmogorov, Wikimedia Commons

Key idea: micro dynamics (drift + noise) $\Rightarrow$ macro density law of motion.

Standalone Example: Diffusion of Particles

Setup (no economics!): Consider a large population of particles, each independently following:

$$dX_t = \mu \, \mathrm{d}t + \sigma \, \mathrm{d}B_t$$

where μ is a constant drift and σ is a constant diffusion coefficient.

Question: If at time $t = 0$ the particles have density $g_0(x)$, what is the density $g(x, t)$ at time $t > 0$?

Kolmogorov Forward Equation (constant coefficients)

$$\boxed{\frac{\partial g}{\partial t}(x, t) = -\mu \frac{\partial g}{\partial x}(x, t) + \frac{\sigma^2}{2} \frac{\partial^2 g}{\partial x^2}(x, t)}$$

Two competing effects:

- $-\mu \, \partial_x g$: advection — drift transports the density
- $\frac{\sigma^2}{2} \partial_{xx} g$: diffusion — noise spreads the density

Deriving the KFE from First Principles

Goal: Derive $\partial_t g = -\mu \partial_x g + \frac{\sigma^2}{2} \partial_{xx} g$ from basic probability.
Step 1 — For any smooth test function $\varphi(x)$: $\mathbb{E}[\varphi(X_t)] = \int_{-\infty}^{\infty} \varphi(x)\, g(x,t)\, \mathrm{d}x$
Step 2 — Differentiate w.r.t. t and use Itô's lemma on $\varphi(X_t)$:

$$\frac{d}{dt} \int \varphi\, g\, \mathrm{d}x = \mathbb{E}\Big[\mu\, \varphi'(X_t) + \tfrac{\sigma^2}{2} \varphi''(X_t)\Big] = \int \Big[\mu\, \varphi' + \tfrac{\sigma^2}{2} \varphi''\Big]\, g\, \mathrm{d}x$$

Step 3 — Integrate by parts (assuming $g \to 0$ at $\pm\infty$):

$$\int \mu\, \varphi'\, g\, \mathrm{d}x = -\int \varphi\, \mu\, \partial_x g\, \mathrm{d}x \qquad \text{(once)}$$

$$\int \tfrac{\sigma^2}{2} \varphi''\, g\, \mathrm{d}x = \int \varphi\, \tfrac{\sigma^2}{2} \partial_{xx} g\, \mathrm{d}x \qquad \text{(twice)}$$

Step 4 — Since this holds for *all* test functions φ: $\boxed{\partial_t g = -\mu\, \partial_x g + \frac{\sigma^2}{2}\, \partial_{xx} g}$ ✓

The KFE: General Itô Diffusion

For $dX_t = \mu(X_t)\,\mathrm{d}t + \sigma(X_t)\,\mathrm{d}B_t$ with state-dependent drift and diffusion, the density $g(x,t)$ satisfies the **general KFE**:

Kolmogorov Forward / Fokker–Planck Equation

$$\boxed{\frac{\partial g}{\partial t}(x,t) = -\frac{\partial}{\partial x}\left[\mu(x)\,g(x,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left[\sigma(x)^2\,g(x,t)\right]}$$

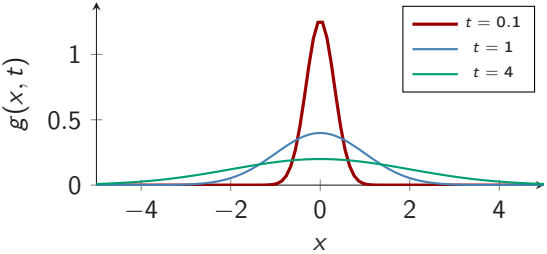
Important: when σ depends on x, it enters *inside* the second derivative:
$\frac{1}{2}\partial_{xx}\left[\sigma(x)^2 g\right] \neq \frac{\sigma(x)^2}{2}\partial_{xx} g$

In divergence form: $\partial_t g = -\partial_x \underbrace{\left[\mu(x)g - \frac{1}{2}\partial_x(\sigma^2 g)\right]}_{=:\text{probability flux } J(x,t)}$ Conservation law: $\partial_t g + \partial_x J = 0$.

Example 1: The Heat Equation (Pure Diffusion)

Setting: No drift ($\mu = 0$), constant diffusion σ. Particles start at x_0: $\partial_t g = \frac{\sigma^2}{2}\partial_{xx} g$, $g(x, 0) = \delta(x - x_0)$.

Solution: Gaussian with spreading variance: $g(x, t) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left(-\frac{(x-x_0)^2}{2\sigma^2 t}\right)$



Physical interpretation:

- Heat spreads from a point source
- Variance grows linearly: $\text{Var} = \sigma^2 t$
- Total "mass" conserved: $\int g = 1$
- Most fundamental PDE in physics

This is exactly what PINNs solve (Day 5)!

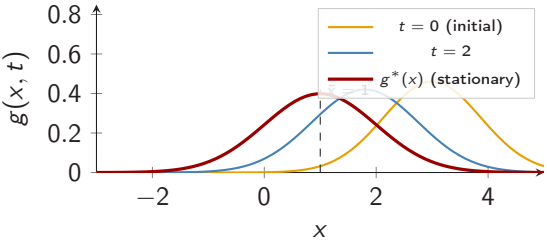
Example 2: Ornstein–Uhlenbeck Process

Setting: $dX_t = \theta(\bar{x} - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$ (mean-reverting).

KFE: $\partial_t g = \theta\,\partial_x[(x - \bar{x})\,g] + \frac{\sigma^2}{2}\partial_{xx} g$

Stationary distribution ($\partial_t g = 0$):

$$\boxed{g^*(x) = \frac{1}{\sqrt{2\pi\sigma^2/(2\theta)}} \exp\left(-\frac{(x-\bar{x})^2}{2\cdot\sigma^2/(2\theta)}\right) = \mathcal{N}\left(\bar{x},\ \frac{\sigma^2}{2\theta}\right)}$$



Interpretation:

- Mean reversion (θ) concentrates mass around $\bar{x}$
- Diffusion (σ) spreads mass
- Balance $\Rightarrow$ Gaussian steady state
- Larger $\theta \Rightarrow$ tighter distribution

This will model aggregate productivity Z_t in the Krusell–Smith model.

Example 3: Boundaries and Borrowing Constraints

Setting: $dX_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}B_t$ on $X_t \geq 0$.

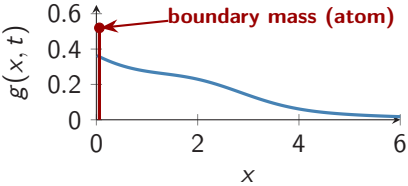
- **Absorbing BC**: mass exits at the boundary.
- **Reflecting BC**: no net probability flow across the boundary.
- In Huggett/Aiyagari (asset drift only), borrowing constraint $a \geq \underline{a}$ is a state constraint.

For the absolutely continuous density part:

$$J(a, n) := s(a, n)\, g_t(a, n), \qquad J(\underline{a}, n) = 0.$$

Boundary atoms can form at $\underline{a}$ when constrained agents pile up.

No-outflow at boundary



From Physics to Economics: A Continuum of Agents

Key insight (propagation of chaos):
Consider a continuum of agents $i \in [0, 1]$, each independently following:

$$dX_t^i = \mu(X_t^i)\,\mathrm{d}t + \sigma(X_t^i)\,\mathrm{d}B_t^i$$

where B_t^i are *independent* Brownian motions (idiosyncratic shocks).
By the **law of large numbers** at the population level:

- The cross-sectional density $g(x, t)$ evolves deterministically
- Even though each individual path is random!
- The density satisfies the KFE: $\partial_t g = -\partial_x[\mu g] + \frac{1}{2}\partial_{xx}[\sigma^2 g]$

Analogy: Each molecule of air moves randomly (Brownian motion), but the *temperature distribution* in a room evolves deterministically (heat equation).

In economics: each household's income is random, but the *wealth distribution* evolves deterministically (KFE).

KFE in Economics: The Wealth Distribution

Setup: Agent i has wealth a_t^i and income state $n_t^i \in \{n_1, n_2\}$.
Individual wealth evolves as:

$$da_t^i = \underbrace{s(a_t^i, n_t^i)}_{\text{savings} = \text{income} - \text{consumption}} \, \mathrm{d}t, \qquad a_t^i \geq \underline{a}$$

Income n_t^i switches between $n_1 \leftrightarrow n_2$ with Poisson intensities λ_1, λ_2.
The cross-sectional density $g_t(a, n)$ satisfies:

$$\boxed{\partial_t g_t(a, n) = -\partial_a[s(a, n)\, g_t(a, n)] - \lambda(n)\, g_t(a, n) + \lambda(\hat{n})\, g_t(a, \hat{n})}$$

where $\hat{n}$ denotes the complementary income state.

- $-\partial_a[s \cdot g]$: flow of agents along the wealth axis (savings)
- $-\lambda(n)\, g(a, n)$: agents leaving state n (exit from $n \to \hat{n}$)
- $+\lambda(\hat{n})\, g(a, \hat{n})$: agents entering state n (entry from $\hat{n} \to n$)

Stationary Wealth Distribution

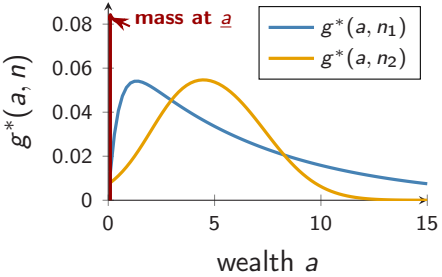
Setting $\partial_t g = 0$, the stationary (ergodic) distribution solves:

$$
\begin{aligned}
0 = & -\partial_a[s(a,n)\, g^*(a,n)] \\
& -\lambda(n)\, g^*(a,n) + \lambda(\hat{n})\, g^*(a,\hat{n})
\end{aligned}
$$

Key features of $g^*(a,n)$:

- Right-skewed: many poor, few rich
- Mass point at borrowing constraint $\underline{a}$ for low-income agents
- Separate densities for each income state
- Normalization: $\sum_n \int_{\underline{a}}^{\infty} g^*(a,n)\, \mathrm{d}a = 1$

This is the object solved numerically in Aiyagari (1994) and Achdou et al. (2022).



Why Does the KFE Not Have a dB_t^0 Term?

Observation: The KFE for $g_t(a, n)$ is a deterministic PDE, even though each agent faces random shocks.

Why?

- Agents face idiosyncratic shocks dB_t^i (independent across i)
- By the **law of large numbers**: randomness washes out at the population level
- $\Rightarrow$ The density g_t evolves deterministically

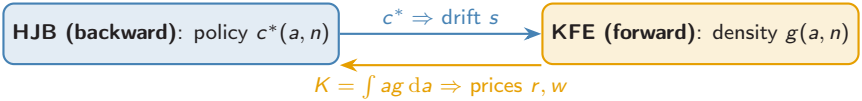
When would dB_t^0 appear in the KFE?

- If there are aggregate shocks (e.g., TFP Z_t following an OU process)
- Then prices r_t, w_t become stochastic $\Rightarrow$ drift is stochastic
- $\Rightarrow$ The density g_t becomes a stochastic process adapted to $\mathcal{F}_t^0$
- This is the Krusell–Smith (1998) setting — much harder!

Aiyagari (no aggregate shocks): KFE is deterministic ✓ **Krusell–Smith** (aggregate shocks): KFE is stochastic $\Rightarrow$ "master equation"

Connection to Mean Field Games

The KFE is one half of the Mean Field Game (MFG) system:



- **Lasry & Lions** (2007): formalized MFGs as coupled PDE systems
- In economics: "Mean Field Game" = competitive equilibrium with continuum of agents
- Each agent is "small" $\Rightarrow$ takes distribution as given
- But collectively, agents *determine* the distribution
- Fixed point: beliefs about g are consistent with optimal behavior

KFE: Summary

Version	KFE	Application
Pure diffusion	$\partial_t g = -\mu \partial_x g + \frac{\sigma^2}{2} \partial_{xx} g$	Heat equation, particle diffusion
General Itô	$\partial_t g = -\partial_x[\mu g] + \frac{1}{2}\partial_{xx}[\sigma^2 g]$	OU process, GBM
With Poisson	$\cdots - \lambda(n)g(n) + \lambda(\hat{n})g(\hat{n})$	Income switching
Economic	drift $= s(a, n)$ (endogenous)	Wealth distribution

Key takeaways:

- The KFE maps $\{\text{drift } \mu, \text{diffusion } \sigma, \text{initial density } g_0\} \to$ density evolution $g(x, t)$
- In economics: drift = savings (endogenous, from HJB)
- Without aggregate shocks: KFE is deterministic (LLN)
- With aggregate shocks: KFE becomes stochastic $\Rightarrow$ "master equation"
- State constraints can require boundary atoms in addition to an interior density

Part III

The Hamilton–Jacobi–Bellman Equation

Stochastic Optimal Control in Continuous Time

The Agent's Optimization Problem

Setup: Agent i with state $x_t^i = (a_t^i, n_t^i)$ chooses consumption c_t^i to solve:

$$\max_{\{c_t^i\}_{t\geq 0}} \mathbb{E}\left[\int_0^\infty e^{-\rho t}\, u(c_t^i)\,\mathrm{d}t\right]$$

subject to:

$$da_t^i = \underbrace{(\tilde{w}_t n_t^i + \tilde{r}_t a_t^i - c_t^i)}_{\equiv\, s(c_t^i,\, a_t^i,\, n_t^i,\, \tilde{w}_t,\, \tilde{r}_t)}\,\mathrm{d}t, \qquad a_t^i \geq \underline{a}$$

$$n_t^i \in \{n_1, n_2\}, \quad \text{switches with Poisson intensity } \lambda(n_t^i)$$

- $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$: CRRA utility, $\gamma > 0$ (risk aversion)
- $\rho > 0$: discount rate
- $(\tilde{r}, \tilde{w})$: agent's beliefs about price processes (rental rate, wage)
- $\underline{a}$: exogenous borrowing limit

The Value Function

Definition

The value function $V(a, n)$ is the maximum expected discounted utility starting from state (a, n):

$$V(a, n) = \max_{\{c_t\}} \mathbb{E}\left[\int_0^\infty e^{-\rho t}\, u(c_t)\, \mathrm{d}t \;\middle|\; a_0 = a,\ n_0 = n\right]$$

Properties of V:

- V is increasing in a (more wealth $\Rightarrow$ higher utility)
- V is concave in a (diminishing marginal value of wealth)
- $V(a, n_2) > V(a, n_1)$ if $n_2 > n_1$ (higher income $\Rightarrow$ higher value)
- $V'(a) = \partial_a V(a, n)$ is the marginal value of wealth

Goal: Find $V(a, n)$ and the optimal policy $c^*(a, n)$. $\Rightarrow$ Use the Hamilton–Jacobi–Bellman equation.

Deriving the HJB: Five Steps

Step 1 — Dynamic Programming Principle. For small $h > 0$:

$$V(a,n) = \max_c \left\{ \int_0^h e^{-\rho t} u(c_t)\,\mathrm{d}t + e^{-\rho h}\,\mathbb{E}[V(a_h, n_h) \mid a_0 = a, n_0 = n] \right\}$$

Step 2 — Itô's lemma on $V(a_t, n_t)$ (between jumps of n):

$$dV = V'(a,n) \cdot s(c,a,n)\,\mathrm{d}t$$

(no diffusion term since $da = s\,\mathrm{d}t$ is deterministic between jumps)
Step 3 — Take expectations, account for Poisson jumps:

$$\mathbb{E}[dV] = \left[V'(a,n) \cdot s + \lambda(n)\big(V(a,\hat{n}) - V(a,n)\big)\right]\,\mathrm{d}t$$

Step 4 — Substitute into DPP, divide by h, let $h \to 0$:
Step 5 — Impose optimality over c:

$$\boxed{\rho V(a,n) = \max_c \left\{ u(c) + V'(a,n) \cdot (wn + ra - c) + \lambda(n)\big(V(a,\hat{n}) - V(a,n)\big) \right\}}$$

The HJB Equation: Interpretation

$$\rho V(a,n) = \max_c \Big\{ u(c) + \underbrace{V'(a,n)\cdot s(a,n,c)}_{\text{marginal value} \times \text{savings}} + \underbrace{\lambda(n)\big(V(a,\hat{n}) - V(a,n)\big)}_{\text{expected change from income switch}} \Big\}$$

Interpretation ("required return on the value function"):

- LHS: ρV = "required return" = discount rate $\times$ asset value
- RHS: "dividend" + "capital gain" = total return
  - $u(c)$: flow utility ("dividend")
  - $V'(a,n)\cdot s$: expected capital gain from saving
  - $\lambda(n)(V(\hat{n}) - V(n))$: expected gain/loss from income switch

The HJB is an asset pricing equation:
the value function is priced like a financial asset.

PAUSE — 5 min Finger Exercise: The HJB Equation

Exercise

A household with wealth a earns income y and chooses consumption c to maximize utility. The HJB equation is:

$$\rho\, V(a) = \max_c \left\{ u(c) + V'(a)\,(y + ra - c) \right\}.$$

1. Take the first-order condition w.r.t. c. What is the relationship between $u'(c)$ and $V'(a)$?
2. With $u(c) = \log(c)$: express optimal c^* in terms of $V'(a)$.
3. Interpret: what does $V'(a)$ represent economically?

Solution: The HJB Equation

Answer

1. FOC: $u'(c^*) = V'(a)$. Marginal utility of consumption equals the marginal value of wealth.
2. With $u(c) = \log(c)$: $1/c^* = V'(a)$, so $c^*(a) = 1/V'(a)$.
3. $V'(a)$ is the shadow price of wealth—the marginal value of an extra unit of assets. When $V'(a)$ is high (wealth is very valuable), consumption is low (the agent saves aggressively).

Optimal Policy: The Consumption Function

First-order condition (differentiate HJB w.r.t. c):

$$u'(c^*) = V'(a,n) \qquad \Longleftrightarrow \qquad \boxed{c^*(a,n) = (u')^{-1}(V'(a,n))}$$

For CRRA utility $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $u'(c) = c^{-\gamma}$:

$$c^*(a,n) = [V'(a,n)]^{-1/\gamma}$$

Savings function:

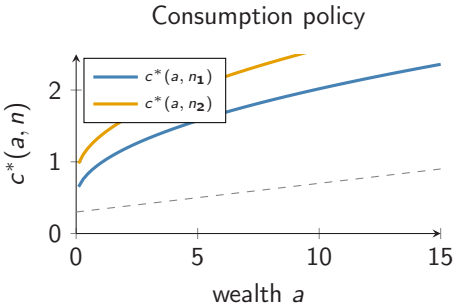
$$s(a,n) = wn + ra - c^*(a,n)$$

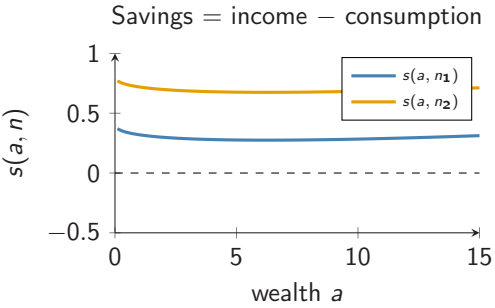
State constraint at $a = \underline{a}$:

$$\begin{aligned} &0 < c^*(\underline{a},n) \leq wn + r\underline{a}, \\ &c^*(\underline{a},n) = \min\{[V_a(\underline{a},n)]^{-1/\gamma},\ wn + r\underline{a}\}, \\ &s(\underline{a},n) = wn + r\underline{a} - c^*(\underline{a},n) \geq 0. \end{aligned}$$

Interior: $u'(c^*) = V_a$. Boundary: constrain consumption so wealth cannot drift below $\underline{a}$.

Consumption and Savings: Qualitative Behavior





- Low-income agents near $\underline{a}$: savings near zero, often constrained
- High-income agents: positive savings $\Rightarrow$ accumulate wealth
- Precautionary savings: more saving than in the certainty benchmark

HJB with Aggregate Shocks (Krusell–Smith)

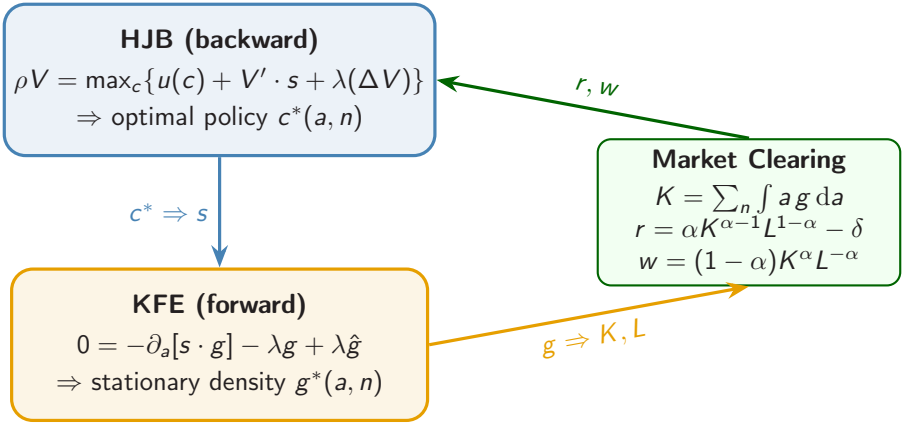
When aggregate TFP Z_t follows $dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma^z\,\mathrm{d}B_t^0$, the value function depends on the entire wealth distribution g: $V = V(a, n, z, g)$. The **HJB** becomes:

$$\begin{aligned}\rho V(a,n,z,g) = \max_c \Big\{ & u(c) + \partial_a V \cdot \big(w(z,g)\,n + r(z,g)\,a - c\big) \\ & + \lambda(n)\big(V(a,\hat{n},z,g) - V(a,n,z,g)\big) \\ & + \partial_z V \cdot \mu^z(z) + \tfrac{1}{2}(\sigma^z)^2\,\partial_{zz} V + \sum_j \int_{\underline{a}}^{\infty} \frac{\delta V}{\delta g_j}(a, n_j, z, g)(y) \cdot \mu_j^g(y,z,g)\,\mathrm{d}y \Big\}\end{aligned}$$

The Curse of Dimensionality

g is infinite-dimensional. Read $\frac{\delta V}{\delta g_j}(\cdot)(y)$ as the Gâteaux/functional derivative: how V changes when a small bump of mass is added at wealth y in productivity-state j. After a finite-dimensional approximation of g (cf. Deck 2 *Three Approximations*), it collapses to ordinary partials $\partial V / \partial \hat{\varphi}_i$. This makes the PDE intractable by standard methods $\Rightarrow$ deep learning (EMINNs — Deck 2).

The Coupled HJB–KFE System



Solution method: iterate until convergence (Aiyagari, no aggregate shocks):

1. Guess $r \to$ solve HJB for $V, c^* \to$ solve KFE for $g^* \to$ compute $K \to$ update r

Part IV

Competitive Equilibrium

Aiyagari (1994) and Krusell–Smith (1998)

Setting: Agents and Preferences

- **Continuous time**, infinite horizon, one consumption good
- **Population**: continuum of agents $i \in I = [0, 1]$
- **Preferences**: $\mathbb{E}\left[\int_0^\infty e^{-\rho t}\, u(c_t^i)\, \mathrm{d}t\right], \quad u(c) = \frac{c^{1-\gamma}}{1-\gamma}$
- **Labor endowment**: $n_t^i \in \{n_1, n_2\}$ follows a CTMC:
  - $n_1 \to n_2$ with Poisson intensity λ_1; $\quad n_2 \to n_1$ with intensity λ_2

Parameter	**Interpretation**
ρ	Discount rate
γ	Risk aversion (inverse EIS)
$n_1 < n_2$	Low / high labor productivity
λ_1, λ_2	Transition intensities

Production Technology

Representative firm: $Y_t = e^{Z_t} F(K_t, L_t) = e^{Z_t} K_t^{\alpha} L_t^{1-\alpha}$

- K_t, L_t: capital and labor hired at time t
- F: CRS production function (Cobb–Douglas)
- **Aiyagari** (1994): $Z_t = \bar{Z}$ constant $\Rightarrow$ no aggregate risk
- **Krusell–Smith** (1998): Z_t follows OU: $dZ_t = \eta(\bar{Z} - Z_t)\,\mathrm{d}t + \sigma^z\,\mathrm{d}B_t^0$

Additional assumptions:

- Consumption $\leftrightarrow$ capital conversion at $1:1$ ratio
- Capital depreciates at rate δ

Markets

- **Incomplete financial markets**:
  - Agents cannot insure idiosyncratic income risk
  - Instead, agents "save and borrow claims on capital"
  - a_t: agent's claim on capital stock
  - Borrowing constraint: $a_t \geq \underline{a} > -\infty$
- **Competitive capital market** with rental rate r_t:

$$\text{Clearing:} \quad \int_I a_t^i \, \mathrm{d}i = K_t$$

- **Competitive labor market** with wage w_t:

$$\text{Clearing:} \quad \int_I n_t^i \, \mathrm{d}i = L_t$$

- **Goods market**: clears by Walras's Law

If $a_t < 0$, interpret this as borrowing from a frictionless intermediary that pools deposits and finances capital.

Firm Problem

The representative firm solves:

$$\max_{K_t, L_t} \left\{ e^{Z_t} F(K_t, L_t) - (r_t + \delta) K_t - w_t L_t \right\}$$

First-order conditions:

$$r_t = e^{Z_t} \partial_K F(K_t, L_t) - \delta = \alpha e^{Z_t} K_t^{\alpha - 1} L_t^{1-\alpha} - \delta$$

$$w_t = e^{Z_t} \partial_L F(K_t, L_t) = (1 - \alpha) e^{Z_t} K_t^{\alpha} L_t^{-\alpha}$$

Prices are functions of the aggregate state:

$$q_t = \begin{bmatrix} r_t \\ w_t \end{bmatrix} = Q(Z_t, g_t)$$

where $K_t = \sum_{j \in \{1,2\}} \int a \, g_t(a, n_j) \, \mathrm{d}a$.

Having explicit expressions for prices in terms of (Z_t, g_t) makes the problem tractable.

Kolmogorov Forward Equation in Equilibrium

Given stochastic processes $\{c^*, \tilde{r}, \tilde{w}\}$, at time $t > 0$, on $(\underline{a}, \infty)$, the density $g_t(a, n)$ satisfies the KFE:

$$\boxed{\partial_t g_t(a, n) = -\partial_a\Big[\big(\tilde{w}_t n + \tilde{r}_t a - c_t^*\big)\, g_t(a, n)\Big] - \lambda(n)\, g_t(a, n) + \lambda(\hat{n})\, g_t(a, \hat{n})}$$

Observations:

- Why does the KFE not have dB_t^0?
  - Idiosyncratic shocks wash out at the population level (LLN)
  - But with aggregate shocks (KS): g_t is stochastic via prices
- $g_t(a, n)$ is still stochastic in equilibrium since r and w depend on $\mathcal{F}_t^0$
- Low-income agents "stuck" at borrowing constraint $\Rightarrow$ boundary atom in g_t
- Technical: need measure-valued solution (Achdou et al., 2022)

Definition: Rational Expectations Equilibrium

Definition (Rational Expectations Equilibrium)

Given initial density g_0, an equilibrium is $\mathcal{F}_t^0$-adapted processes $\{c_t^i, g_t, z_t, q_t := [r_t, w_t]\}$ such that:

1. **Household optimality**: Given price belief $\tilde{q}$, c_t^i solves HJB.
2. **Firm optimality**: $r_t = e^{Z_t}\partial_K F - \delta, \quad w_t = e^{Z_t}\partial_L F$
3. **Market clearing**: $K_t = \sum_j \int a\, g_t(a, n_j)\, \mathrm{d}a, \quad L = \sum_j \int n_j\, g_t(a, n_j)\, \mathrm{d}a$
4. **Belief consistency**: $\tilde{q} = q \quad$ (beliefs = actual prices)

Derivation Map: Model $\Rightarrow$ HJB $\Rightarrow$ KFE $\Rightarrow$ Equilibrium

Step 1 (household control problem):

$$da_t = (wn_t + ra_t - c_t)\,\mathrm{d}t, \qquad n_t \in \{n_1, n_2\} \text{ (Poisson switching).}$$

Step 2 (HJB + FOC):

$$\rho V(a,n) = \max_c \left\{ u(c) + V_a(a,n)\,[wn + ra - c] + \lambda(n)\big(V(a,\hat{n}) - V(a,n)\big) \right\},$$

$$u'(c^*(a,n)) = V_a(a,n), \qquad s(a,n) := wn + ra - c^*(a,n).$$

Step 3 (KFE induced by optimal drift s):

$$0 = -\partial_a[s(a,n)g^*(a,n)] - \lambda(n)g^*(a,n) + \lambda(\hat{n})g^*(a,\hat{n}).$$

Step 4 (general equilibrium closure, Aiyagari):

$$K = \sum_n \int a\, g^*(a,n)\,\mathrm{d}a, \qquad L = \sum_n \int n\, g^*(a,n)\,\mathrm{d}a,$$

$$r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta, \qquad w = (1-\alpha) K^{\alpha} L^{-\alpha}.$$

Fixed point: iterate on (r, w) until household supply and firm demand are consistent.

Three Notions of Equilibrium

Following Sargent (1999, 2008):

1. **Rational expectations equilibrium**
   - Agents' perceived law of motion = true law of motion
   - Even for data that would *never* be generated on the equilibrium path
2. **Self-confirming belief equilibrium**
   - Agents' model is the best statistical fit to data generated by the economy
   - Agents would never reject their model based on observed data
   - But may have incorrect off-equilibrium beliefs
3. **Optimal misspecified belief equilibrium**
   - Agents use an optimally chosen (but possibly wrong) model
   - Robust decision-making under model uncertainty

In this course: we focus on rational expectations. But note: neural network "solvers" can naturally accommodate bounded rationality (approximate equilibria).

Huggett (1993): Economic Mechanism

Environment

- Endowment economy with idiosyncratic income $y_t \in \{y_1, y_2\}$.
- Incomplete markets: only one risk-free bond b_t.
- Borrowing limit $b_t \geq \underline{b}$.
- Bond is in zero net supply.

Household trade-off

- Low income state: wants to borrow to smooth consumption.
- High income state: saves for self-insurance.
- Constraint creates a mass of constrained agents at $\underline{b}$.

How equilibrium price is determined

- For candidate return q, solve HJB $\Rightarrow$ policy $s(b, y)$.
- Propagate policy through KFE $\Rightarrow$ stationary density $g^*(b, y)$.
- Compute aggregate bond demand:

$$B^s(q) = \sum_y \int_{\underline{b}}^{\infty} b\, g^*(b, y)\, \mathrm{d}b.$$

- Equilibrium return: q^* such that $B^s(q^*) = 0$.

Implication: the risk-free rate is pinned down by precautionary self-insurance demand, not by a production FOC.

Aiyagari (1994): Economic Mechanism I

What changes vs. Huggett

- Add a representative firm with Cobb–Douglas production.
- Household asset is now a claim on aggregate capital a_t.
- Same incomplete insurance and borrowing constraint logic.
- Aggregate capital is endogenously positive: $K > 0$.

Supply side (households)

- For given (r, w), HJB gives policy $s(a, z)$.
- KFE gives density $g^*(a, z)$.
- Capital supplied by households: $K^s(r) = \sum_z \int_{\underline{a}}^{\infty} a\, g^*(a, z)\, \mathrm{d}a$.

Intuition

- Low productivity states generate demand for insurance via saving.
- High productivity states accumulate assets.
- With many uninsured households, aggregate saving pressure is high.
- This pushes the equilibrium return on capital downward.

Aiyagari (1994): Economic Mechanism II

General-equilibrium closure

Firm demand for capital

$$r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta$$

$$w = (1-\alpha) K^{\alpha} L^{-\alpha}$$

These conditions imply a decreasing capital-demand schedule $K^d(r)$.

Household capital supply

$$K^s(r) = \sum_z \int_{\underline{a}}^{\infty} a\, g^*(a,z)\, \mathrm{d}a,$$

where g^* is the stationary density induced by optimal savings.

Equilibrium condition

$$K^s(r^*) = K^d(r^*), \qquad w^* = w(r^*).$$

Economic interpretation

Aiyagari is a full GE model: prices are jointly determined by

- household self-insurance behavior (asset supply), and
- firm marginal productivity (asset demand).

Huggett (1993): Formal Stationary PDE System

Primitives: endowment economy, $b_t \geq \underline{b}$, $y_t \in \{y_1, y_2\}$ with Poisson switching. Bonds pay instantaneous return r (price-vs-return convention: r, not a bond price q).

$$db_t = (r\, b_t + y_t - c_t)\,\mathrm{d}t, \qquad \max_{\{c_t\}} \mathbb{E}[]\left[\int_0^\infty e^{-\rho t} u(c_t)\,\mathrm{d}t\right].$$

HJB (state y):

$$\rho V = \max_c \{u(c) + V_b(rb + y - c) + \lambda(y)(V(b, \hat{y}) - V(b, y))\}, \quad u'(c^*) = V_b, \quad s = rb + y - c^*.$$

KFE on $(\underline{b}, \infty)$:

$$0 = -\partial_b[s(b, y)g^*(b, y)] - \lambda(y)g^*(b, y) + \lambda(\hat{y})g^*(b, \hat{y}),$$

$$J(\underline{b}, y) = s(\underline{b}, y)g^*(\underline{b}, y) = 0, \qquad \sum_y \int_{\underline{b}}^\infty g^*(b, y)\,\mathrm{d}b = 1.$$

Bond market clearing:

$$\sum_y \int_{\underline{b}}^\infty b\, g^*(b, y)\,\mathrm{d}b = 0.$$

Unknowns: (V, g^*, q).

Aiyagari (1994): Formal Stationary PDE System

Production side (no aggregate shocks):

$$Y = K^{\alpha} L^{1-\alpha}, \qquad r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta, \qquad w = (1-\alpha) K^{\alpha} L^{-\alpha}.$$

Household problem:

$$\max_{\{c_t\}} \mathbb{E}[] \left[\int_0^{\infty} e^{-\rho t} u(c_t)\, \mathrm{d}t \right], \qquad da_t = (wz_t + ra_t - c_t)\, \mathrm{d}t, \qquad a_t \geq \underline{a}, \quad z_t \in \{z_1, z_2\}.$$

HJB:

$$\rho V(a,z) = \max_c \left\{ u(c) + V_a(a,z)\,[wz + ra - c] + \lambda(z)\big(V(a,\hat{z}) - V(a,z)\big) \right\},$$

$$u'(c^*) = V_a, \qquad s(a,z) := wz + ra - c^*(a,z).$$

KFE on $(\underline{a}, \infty)$:

$$0 = -\partial_a\big[s(a,z) g^*(a,z)\big] - \lambda(z) g^*(a,z) + \lambda(\hat{z}) g^*(a,\hat{z}).$$

Market clearing & normalization:

$$K = \sum_z \int_{\underline{a}}^{\infty} a\, g^*(a,z)\, \mathrm{d}a, \qquad L = \sum_z \int_{\underline{a}}^{\infty} z\, g^*(a,z)\, \mathrm{d}a, \qquad \sum_z \int_{\underline{a}}^{\infty} g^*(a,z)\, \mathrm{d}a = 1.$$

Unknowns: (V, g^*, r, w, K) (or (V, g^*, r) with $w = w(r)$).

Huggett vs. Aiyagari: Key Differences

	Huggett (1993)	**Aiyagari (1994)**
Economy	Endowment	Production ($Y = K^\alpha L^{1-\alpha}$)
Asset	Bond b	Capital claim a
Supply	Zero net ($\int b = 0$)	Positive ($\int a = K > 0$)
Price	Bond return q	Interest rate r, wage w
Clearing	$\int b\, g\, \mathrm{d}b = 0$	$\int a\, g\, \mathrm{d}a = K(r)$

How the distributions differ:

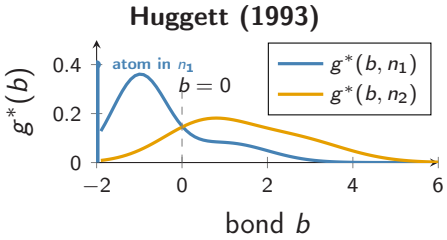
Huggett:

- Wealth centered around 0 (zero net supply)
- Large mass at borrowing constraint
- Symmetric-ish with constrained left tail

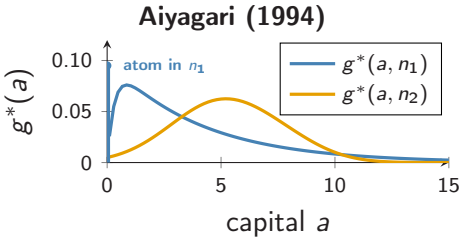
Aiyagari:

- Wealth shifted right ($K > 0$)
- Long right tail (wealthy agents)
- Smaller mass at constraint

Wealth Distributions: Huggett vs. Aiyagari



Net supply $= 0$: mean wealth ≈ 0.



$K > 0$: mean wealth is positive with a right tail.

Why This Matters: HANK and Modern Macro

The Aiyagari/Krusell–Smith framework is the backbone of HANK:

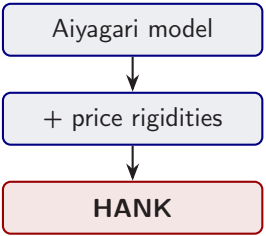
Heterogeneous Agent New Keynesian:

- Kaplan, Moll & Violante (2018, *AER*)
- Replace representative agent with Aiyagari economy
- Add nominal rigidities
- Result: monetary policy affects consumption heterogeneously

Applications:

- Fiscal stimulus (who spends?)
- Inequality dynamics
- Wealth taxation
- COVID-19 policy responses

Key insight from HANK:



"These building blocks are essential for modern macro."

Part V

The Master Equation

From Coupled System to Single PDE

Recursive Representation of Equilibrium

States: individual $x = (a, n)$, aggregate (z, g); value function $V(x, z, g)$.
Given a belief drift $\hat{\mu}^g$, the household HJBE is

$$0 = \max_c \Big\{ -\rho V + u(c) + V_a \left[w(z,g)n + r(z,g)a - c \right] + \lambda(n)\big(V(a, \hat{n}, z, g) - V(a, n, z, g)\big)$$
$$+ V_z\, \mu^z(z) + \tfrac{1}{2}(\sigma^z)^2 V_{zz} + \sum_{j \in \{1,2\}} \int_{\underline{a}}^{\infty} \frac{\delta V}{\delta g_j}(a, n, z, g)(y)\, \hat{\mu}_j^g(y, z, g)\, \mathrm{d}y \Big\}.$$

For $c^*(a, n, z, g; \hat{\mu}^g)$, the density drift is

$$\mu_j^g(a, z, g) = -\partial_a\big[(w(z,g)n_j + r(z,g)a - c_j^*(a, z, g))\, g_j(a)\big] - \lambda_j g_j(a) + \lambda_{\hat{j}} g_{\hat{j}}(a).$$

Equilibrium consistency: $\hat{\mu}^g = \mu^g$.

The Master Equation

Key idea: Substitute KFE, market clearing, and belief consistency into the HJBE. Result: one PDE characterizing the equilibrium value function:

Master Equation

$$
\begin{aligned}
0 = -\rho V(a,n,z,g) + u(c^*) + V_a\left[w(z,g)n + r(z,g)a - c^*\right] \\
+ \lambda(n)(V(a,\hat{n},z,g) - V(a,n,z,g)) + V_z\,\mu^z(z) + \tfrac{1}{2}(\sigma^z)^2 V_{zz} \\
+ \sum_{j\in\{1,2\}} \int_{\underline{a}}^{\infty} \frac{\delta V}{\delta g_j}(a,n,z,g)(y)\,\mu_j^g(y,z,g)\,\mathrm{d}y \;=:\; \mathcal{L}V.
\end{aligned}
$$

where c^* satisfies $u'(c^*) = \partial_a V(a,n,z,g)$.

"One equation to rule them all"

HJB + KFE + market clearing = master equation

The Envelope Condition: Working with $W := \partial_a V$

Practical insight (Gu, Lauriere, Merkel & Payne, 2024): It is easier to work with $W(a, l, z, g) := \partial_a V(a, l, z, g)$ rather than V itself.
Why? Because W directly gives the consumption policy:
$u'(c^*(a, n, z, g)) = W(a, n, z, g) \;\Leftrightarrow\; c^* = W^{-1/\gamma}$
The **master equation for** W (using the envelope theorem):

$$\boxed{0 = (r(z, g) - \rho)\, W + \underbrace{\mathcal{L}_x W}_{\text{individual}} + \underbrace{\mathcal{L}_z W}_{\text{aggregate TFP}} + \underbrace{\mathcal{L}_g W}_{\text{distribution}} =: \mathcal{L} W}$$

where:

- $\mathcal{L}_x W$: individual state dynamics (savings, income switching)
- $\mathcal{L}_z W$: aggregate TFP dynamics ($\partial_z W \cdot \mu^z + \frac{1}{2}(\sigma^z)^2 \partial_{zz} W$)
- $\mathcal{L}_g W$: distribution dynamics (functional derivative w.r.t. g)

Why the Master Equation?

Advantages:

- ✓ Single PDE instead of coupled HJB + KFE + MC
- ✓ No separate market clearing step
- ✓ Amenable to neural network approximation: parameterize V or W by a deep net
- ✓ Auto-differentiation computes all derivatives (including functional derivative w.r.t. g)

Challenges:

- ✗ Infinite-dimensional: functional derivative $\frac{\delta V}{\delta g_j}$
- ✗ Need finite-dimensional approximation of g
- ✗ Training can be unstable ("bad approximate solutions")
- ✗ No convergence guarantees

Companion (numerical) deck: three approaches to approximate g in finite dimensions:
(1) Finite population, (2) Discrete state space, (3) Projection onto basis functions.
Then: solve the master equation with EMINNs (neural networks + auto-diff).

Summary and Key Takeaways

Stochastic Calculus

Brownian motion, Itô processes, Itô's lemma, Poisson jumps — the language of continuous-time models.

KFE / Fokker–Planck

Tracks density evolution of a stochastic process. In economics: the wealth distribution.

HJB + Equilibrium

Optimal control $\Rightarrow$ HJB. Coupled with KFE via savings policy. Master equation unifies all.

Next (numerical methods):

1. Solve the coupled HJB+KFE by finite differences (benchmark)
2. Solve the same system with PINNs (mesh-free, auto-diff)
3. Solve the master equation with EMINNs (deep learning)

End of theory.
The numerical methods deck picks up here.

From Theory to Code (Companion Notebook)

Companion notebook:

- `lecture_13_B11_continuous_time_ha_numerics/code/lecture_13_B11_08_Aiyagari_Continuous_Time_FD_and_PINN_PyTorch.ipynb`

Covers:

- Finite-difference solver for the Aiyagari HJB+KFE system
- PINN solver for the same equilibrium
- Comparison of stationary distributions, savings policies, and the equilibrium interest rate

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Questions?

Simon Scheidegger

HEC, University of Lausanne

https://sischei.github.io

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Covered in this deck:

I Stochastic Calculus
II Kolmogorov Forward Eq.
III Hamilton–Jacobi–Bellman
IV Competitive Equilibrium
V The Master Equation

Next:

Numerical Methods
FD $\rightarrow$ PINNs $\rightarrow$ EMINNs